

QF600 Asset Pricing — Week 4 Reference Cheatsheet

Linear Factor Models: K-factor model, Fama-MacBeth, Fama-French, Carhart momentum, the factor zoo

Built from Lecture 4 slides (Liu Peng, LKCSB) · September 2026

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Source: “Lecture slides – Week 4 v3” (Liu Peng, LKCSB, 38 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged **[Beyond slides]**. △ marks a slide ambiguity or error, with the corrected version.

Exam context (Week 1 [S5]): closed book, one double-sided A4 sheet, 40% MCQ and 60% written (concepts plus simple calculations), **no coding**. Python here supports the coding session [S38]. In the exam you need the model equations, how to read loadings, how factors are built, the Fama–MacBeth steps, and hand arithmetic for expected returns and alphas.

Notation used throughout (slide conventions)

Symbol	Meaning	Dim / units	Time index	Notes
$\tilde{R}_i$	return on asset i	scalar, % or decimal per period	per period t (written $\tilde{R}_{it}$ in [S12])	Tilde = random variable
R_f (RF)	risk-free rate (one-month T-bill [S21])	per period	known at the start of the period	Not random
$\tilde{R}_i - R_f$	excess return of asset i	per period	t	The left-hand side of every model this week
$\tilde{R}_M$ (also $\tilde{R}_m$)	market return (value-weighted CRSP)	per period	t	The slides use both M and m
K / k	number of factors / factor index $k = 1, \dots, K$	integer	—	△ [S5] writes the last term as $\beta_{ik}\tilde{F}_k$ (and says “k-factor model”). It should read $\beta_{iK}\tilde{F}_K$.
$\tilde{F}_k$	return on factor k (a traded, excess / long-short return)	per period	t	Observed , not estimated [S9]
α_i	alpha: expected excess return the factors leave unexplained	per period	constant	Model prediction: $\alpha_i = 0$
β_{ik}	loading (exposure) of asset i on factor k	dimensionless	constant over the sample	Estimated by a time-series regression [S9]
$\tilde{\varepsilon}_i$	idiosyncratic error	per period	t	$E = 0$; uncorrelated with the factors and with other assets' ε [S7]
λ_k	risk premium (price of risk) of factor k	per period (return per unit β)	constant	Traded factor: $\lambda_k = E[\tilde{F}_k]$ [S9, S11]
λ_M , MRP	market risk premium $E[\tilde{R}_M - R_f]$	per period	—	MRP = “market risk premium”

Symbol	Meaning	Dim / units	Time index	Notes
$\beta_i, \gamma_i, \delta_i, \zeta_i$	loadings on Mkt-RF, SMB, HML, UMD in FF3 / Carhart	dimensionless	—	⚠ γ_i (a size loading) is not the Fama-MacBeth slope $\tilde{\gamma}_{1t}$
$\tilde{\gamma}_{0t}, \tilde{\gamma}_{1t}$	Fama-MacBeth cross-sectional intercept and slope at date t	per period	t	
$\tilde{\eta}_{it}$	cross-sectional regression error	per period	t	
T, N	number of periods, number of assets	integers	—	
SMB, HML, UMD (WML, Mom), RMW, CMA	factor-mimicking long-short portfolio returns	per period	t	Definitions in 1.11 and 1.13
S/L, S/M, ..., B/H	the six size × B/M portfolios	returns	t	S = small, B = big; L/M/H = low/medium/high B/M
B/M	book-to-market ratio (book equity / market equity)	ratio	sorting date	

Units trap: the Ken French files are in **percent per month**. Divide by 100 before compounding or mixing with decimals [S19-S20].

0. Topic map

PREREQUISITES

- W1 Expected utility, risk aversion → mean-variance preferences
- W2 Frontier, tangency portfolio, estimators $\hat{\mu} = (1/T)\sum r_t$, σ^2
- W3 CAPM $\mu_i = R_f + \beta_i(\mu_M - R_f)$; regression form $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \tilde{\epsilon}_i$;
 α = Jensen's alpha (tested $H_0: \alpha = 0$); β = market sensitivity; SML
 | "One factor left too much unexplained" [S4]



WEEK 4: LINEAR FACTOR MODELS

- Motivation [S2-S3]: Aug 2007 quant crash – 4 funds, 1 shared factor bet
- K-factor model [S5, S7]: $\tilde{R}_i - R_f = \alpha_i + \sum_k \beta_{ik} \tilde{F}_k + \tilde{\epsilon}_i$ (CAPM = K=1)
 | R^2 rises even on random regressors → overfitting warning [S5]
- ACT 1 More than one risk [S6-S15]
 - Geometry: line → plane → hyperplane; $E[R]$ linear in loadings [S8]
 - $\tilde{F}$ (observed) → β (time-series regression) → $\lambda = E[\tilde{F}]$ (traded) [S9]
 - Pricing equation $E[\tilde{R}_i] = R_f + \sum_k \beta_{ik} \lambda_k$; loading = asset's, premium = economy's [S10]
 - λ_M = MRP; CAPM is $k = 1$; $\lambda \approx 0 \Rightarrow$ unpriced [S11]
 - Fama-MacBeth: TS pass → CS pass each $t \rightarrow$ average γ_{1t} ; SE from time variation [S12]
 - Factors are self-financing long-short portfolios; indicative premia [S13]
 - Worked two-factor example: 6.8% [S14] · Quiz Q1-Q3 [S15]
- ACT 2 Fama and French [S16-S26]
 - Size (Banz 1981) & value anomalies → build them as factors [S17]
 - FF3: $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \gamma_i \text{SMB} + \delta_i \text{HML} + \tilde{\epsilon}_i$ [S18]
 - Data: Ken French library, % units, -99.99, Mkt-RF already excess [S19-S20]
 - Construction: Mkt-RF [S21]; 2×3 sorts → SMB [S22], HML [S23]

- Style map: γ vs 6 for ten industries recovers "growth/value/small/large" [S24]
- Carhart = FF3 + $\zeta \cdot \text{UMD}$; "much of what looked like skill was momentum" [S25] · Quiz Q4–Q6 [S26]
- ACT 3 Momentum and beyond [S27–S38]
 - Momentum pattern: winners keep winning ~1 year, reverse at 3–5 years [S28]
 - Carhart four-factor; classroom UMD stand-in (industries, 12-1) [S29]
 - Jegadeesh–Titman construction: deciles, EW, J/K = 3–12 months [S30]
 - Momentum loadings across industries (Energy +0.53, Durables –0.58) [S31]
 - Factor-zoo defence, four tests [S32–S36]:
 - 1 out of sample / $t > 3$ 2 economic story ($\lambda \neq 0$) 3 spanning α 4 net of costs
 - Quiz Q7–Q9 [S37]; coding session [S38]
- ∇
- LATER: W5 portfolio construction uses $\tilde{r} = \alpha + B\tilde{f} + \tilde{\varepsilon}$, $\Sigma = B\Omega B^T + D$, factor neutrality;
- W6 SDF: linear factor models $\Leftrightarrow$ SDF linear in factors

Dependency logic:

- **Time-series regressions** give the β s (asset by asset).
- For **traded** factors, the premium is the time-series mean of the factor. The **cross-section** (Fama–MacBeth) is how you *test or estimate* whether a characteristic's exposure is priced.
- **FF3 and Carhart** are specific choices of which traded factors go into the K -factor model.
- **Act 3** asks whether each new factor deserves a column at all.

1. Concept entries

1.1 Motivation: hidden shared factor exposure (August 2007) [S2–S4]

- **Facts (slide):**
 - In one week of August 2007, the S&P 500 rose **+1.4%** over the whole week (an ordinary week).
 - A simulated daily contrarian long/short quant strategy (Khandani & Lo 2011, unleveraged) lost **–6.85% in just three days**, "a loss the models said could never happen".
 - The fund types involved were a stat-arb fund, a market-neutral fund, a quant value fund and a multi-strat desk. All four fed into **one shared factor exposure**.
- **Lesson [S3]:**
 - "4 funds $\rightarrow$ 1 bet". What looks like four managers with four separate mandates is "mostly the market, tilted to size and value, short momentum; every fund leans the same way".
 - A factor is a **shared risk you may already hold without knowing**. This week names it, measures it and prices it.
- **Transition [S4]:** "One factor left too much unexplained." The CAPM's single market factor fails to explain the cross-section of returns (size, value and momentum anomalies).
- **Intuition:**
 - Diversifying across managers diversifies *idiosyncratic* risk, not *common factor* risk.
 - When many funds load on the same factor, a shock to it (or forced deleveraging) hits all of them at once. Correlations jump in a crisis.
- **[Beyond slides]:** Khandani–Lo's explanation is a liquidity/deleveraging cascade in crowded quant equity strategies (7–9 August 2007). This matches the slide's "one shared factor exposure".
- **Trap:** "market-neutral" $\neq$ "factor-neutral". A book can have zero market beta and still carry large size/value/momentum loadings (connects to W5 factor neutrality, $B^T w = 0$).
- **Connections:** W3 (CAPM's single factor); W5 (factor-neutral portfolios).

1.2 The K-factor model (time-series / return-generating form) [S5, S7]

- **Definition:**

$$\tilde{R}_i - R_f = \alpha_i + \beta_{i1} \tilde{F}_1 + \beta_{i2} \tilde{F}_2 + \cdots + \beta_{iK} \tilde{F}_K + \tilde{\varepsilon}_i = \alpha_i + \sum_{k=1}^K \beta_{ik} \tilde{F}_k + \tilde{\varepsilon}_i.$$

- $\tilde{R}_i - R_f$ is the **excess return**: asset i 's return above the risk-free rate, “the reward it must earn”.
- α_i is **alpha**: the piece of return the factors leave unexplained (abnormal gain or mispricing).
- β_{ik} are the **factor loadings**: how strongly asset i responds to each common factor.
- $\tilde{F}_k$ are the **common factors**: the systematic forces that move returns across all assets.
- $\tilde{\varepsilon}_i$ is the **idiosyncratic error**: firm-specific risk that averages zero and diversifies away.
- **Three pieces [S7]:**
 - **Factors are shared**: every asset sees the same $\tilde{F}_k$; only the loadings β_{ik} differ. That shared part is the common risk.
 - ε is **yours alone**: it is uncorrelated with the factors *and* with every other asset's ε , which is what makes it diversifiable.
 - **CAPM is the $K = 1$ case**: one factor, the market. “Everything from Week 3 still holds.”
 - **Only the middle piece is rewarded**: α_i is a constant and ε_i diversifies away.
- **Assumptions:**
 - $E[\tilde{\varepsilon}_i] = 0$.
 - $\text{Cov}(\tilde{\varepsilon}_i, \tilde{F}_k) = 0$ for all k .
 - $\text{Cov}(\tilde{\varepsilon}_i, \tilde{\varepsilon}_j) = 0$ for $i \neq j$ (the “strict” factor model) [S7].
 - Loadings are constant over the estimation window [Beyond slides].
 - If residuals are correlated across assets, there is an **omitted factor**. Residual risk then does not diversify away, and β estimates may be biased.
- **Implied moments [Beyond slides]:**

$$E[\tilde{R}_i - R_f] = \alpha_i + \sum_k \beta_{ik} E[\tilde{F}_k], \quad \text{Var}(\tilde{R}_i) = \beta_i^\top \Omega \beta_i + \sigma_{\varepsilon_i}^2, \quad \text{Cov}(\tilde{R}_i, \tilde{R}_j) = \beta_i^\top \Omega \beta_j.$$
 - $\Omega = \text{Cov}(\tilde{F})$.
 - Stacked over assets this is W5's $\Sigma = B\Omega B^\top + D$.
- **Interpretation:**
 - The model splits **risk** (common vs idiosyncratic).
 - With $\alpha_i = 0$ it also becomes a **pricing** statement (1.5).
- **R^2 warning ([S5] footer):** “ R^2 rising on columns of pure random numbers.”
 - In-sample R^2 **never falls** when you add a regressor [Beyond slides proof: OLS minimises the SSR, and setting the new coefficient to 0 reproduces the old fit, so $\text{SSR}_{\text{new}} \leq \text{SSR}_{\text{old}}$].
 - A higher R^2 from more factors is not evidence by itself. Use adjusted $R^2 = 1 - (1 - R^2) \frac{n-1}{n-K-1}$, out-of-sample tests, and the Act 3 checklist [Beyond slides].
 - *Example*: $n = 60$ months. CAPM gives $R^2 = 0.30$, so $\text{adj} = 0.288$. A 4-factor model gives $R^2 = 0.33$, so $\text{adj} = \mathbf{0.281}$. The extra three factors don't pay for themselves.
- **Traps:**
 - Forgetting that the LHS is an **excess** return.
 - Calling α “risk”. It is the *unexplained mean*.
 - Assuming “more factors = better model”.
- **Connections:** W3 CAPM regression; W5 $\tilde{r} = \alpha + B\tilde{f} + \tilde{\varepsilon}$ (same model in matrix form).

1.3 Geometry: one factor is a line, two factors a plane [S8]

- **Statement:** adding a factor adds a **dimension, not a curve**. Expected return stays **linear in the loadings**.
 - One factor: $E[R]$ vs β_{market} is a line (the SML). The slide's line has intercept $\approx 1.5\%$ and rises about 7.5% per unit β , with points between $\beta \approx 0.45$ and 1.5 (read from the chart).

- Two factors: $E[R]$ over $(\beta_{\text{market}}, \gamma_{\text{size}})$ is a plane.
- With K factors you fit a **hyperplane**, and “every asset is supposed to lie on it”.
- **Formula:** $E[\tilde{R}_i] = R_f + \sum_k \beta_{ik} \lambda_k$, a hyperplane in $(\beta_{i1}, \dots, \beta_{iK})$ space with intercept R_f and slopes λ_k .
- **Interpretation:**
 - The vertical distance of an asset from the hyperplane is its α_i .
 - A factor with $\lambda_k = 0$ gives a flat direction, i.e. that exposure is not priced.
- **Trap:** linear in *loadings*, not in characteristics. Size or B/M enter only through the returns of the SMB/HML portfolios the asset co-moves with.

1.4 The three symbols: $\tilde{F}$, β , λ [S9]

Step	Symbol	What it is	How obtained	Status
1	$\tilde{F}_k$	factor return: a traded factor's excess return each period (e.g. market, value)	read from data	Observed
2	β_{ik}	loading	time-series regression of $\tilde{R}_i$ (excess) on $\tilde{F}_k$, one asset over time; the slope is β_{ik}	Estimated
3	λ_k	premium	$\lambda_k = E[\tilde{F}_k]$, the factor's own average return; “no cross-section needed”	Observed (sample mean)

- “Each factor is a tradeable portfolio, so its risk premium is simply its average return: $\lambda_k = E[\tilde{F}_k]$.”
- **Why $\lambda_k = E[\tilde{F}_k]$ for traded factors [Beyond slides derivation]:**
 - Apply the pricing equation to the factor portfolio itself.
 - Regressing $\tilde{F}_k$ on all factors gives loading 1 on itself, 0 on the others, and $\alpha = 0$.
 - So $E[\tilde{F}_k] = 1 \cdot \lambda_k$.
 - (Since $\tilde{F}_k$ is already an excess / zero-cost return, there is no R_f term.)
- **Time-series regression (first pass) [Beyond slides detail]:**
 - Model: $\tilde{R}_{it} - R_{ft} = \alpha_i + \beta_i^\top \tilde{F}_t + \tilde{\varepsilon}_{it}$ for $t = 1, \dots, T$.
 - One factor: $\hat{\beta}_i = \frac{\widehat{\text{Cov}}(R_i - R_f, F)}{\widehat{\text{Var}}(F)}$ and $\hat{\alpha}_i = \overline{R_i - R_f} - \hat{\beta}_i \bar{F}$.
- **Worked example [Beyond slides]** (5 months of excess returns, in %):
 - Data: $F = (2, -1, 3, 0, 1)$ and $R_i - R_f = (2.5, -1.0, 3.8, 0.3, 1.4)$.
 - Means: $\bar{F} = 1.0$ and $\bar{R} = 1.4$.
 - Deviations: $F : (1, -2, 2, -1, 0)$ and $R : (1.1, -2.4, 2.4, -1.1, 0)$.
 - $\widehat{\text{Cov}} = \frac{1.1+4.8+4.8+1.1+0}{4} = 2.95$ and $\widehat{\text{Var}}(F) = \frac{1+4+4+1+0}{4} = 2.5$.
 - $\hat{\beta} = 1.18$ and $\hat{\alpha} = 1.4 - 1.18(1.0) = 0.22\%$ **per month**.
- **Traps:**
 - Regressing *raw* returns on factors: the intercept then absorbs $R_f(1 - \beta)$ and is not α .
 - Treating λ as asset-specific.
 - Using a cross-sectional regression to get β (wrong pass).
- **⚠ Tension with [S10]:**

- [S9] says λ needs “no cross-section”; [S10] says you “estimate it from the cross-section of all assets”.
- Resolution: for **traded** factors, the model *implies* $\lambda_k = E[\tilde{F}_k]$ (time-series mean).
- The cross-section (Fama-MacBeth, 1.6) is needed for **non-traded** factors (e.g. GDP growth, inflation), and it is how you **test** whether the cross-section of average returns actually pays λ per unit β .
- If the model holds, both estimates agree.

- **Implementation:**

```
import numpy as np
def ts_regress(y, X):
    """OLS of excess returns y (T,) on factors X (T,K) with intercept -> alpha, betas, t-stats,
    & R2."""
    Xc = np.column_stack([np.ones(len(y)), X])
    coef, *_ = np.linalg.lstsq(Xc, y, rcond=None)
    e = y - Xc @ coef; n, p = Xc.shape
    se = np.sqrt(np.diag(e @ e / (n - p) * np.linalg.inv(Xc.T @ Xc)))
    r2 = 1 - e @ e / ((y - y.mean()) @ (y - y.mean()))
    return coef[0], coef[1:], coef / se, r2
print(ts_regress(np.array([2.5, -1.0, 3.8, .3, 1.4]), np.array([2, -1, 3, 0, 1]))) # alpha
& 0.22, beta 1.18
```

Pitfall: these are homoskedastic OLS standard errors. Use Newey-West / HAC for serially correlated or heteroskedastic monthly returns.

1.5 Loadings vs premia and the pricing equation; the market risk premium [S10-S11]

- **Definition:** every factor model is “loadings times prices, added up”:

$$E[\tilde{R}_i] = R_f + \beta_{i1}\lambda_1 + \dots + \beta_{iK}\lambda_K \iff E[\tilde{R}_i - R_f] = \sum_k \beta_{ik}\lambda_k \quad (\alpha_i = 0).$$

- **Anatomy [S10]:**

- β_{ik} , **the loading, belongs to the ASSET**. It is how much of this factor this particular asset carries, estimated from that asset’s history. Two assets can load on the same factor very differently.
- λ_k , **the premium, belongs to the ECONOMY**. It is the price the market pays for one unit of this risk, and the same number appears in every asset’s equation.
- “Two assets can face the same factor and load on it completely differently, but they face the same price for it.”

- **Price of risk [S11]:**

- $\lambda_M = E[\tilde{R}_M - R_f]$ is the **market risk premium** (MRP): the average reward for holding equities over cash. For any traded factor, $\lambda_k = E[\tilde{F}_k]$.
- **CAPM is $k = 1$** : set the one factor to Mkt-RF and $E[\tilde{R}_i] = R_f + \beta_i \cdot \text{MRP}$, exactly the CAPM. Extra factors only add more $\beta_{ik}\lambda_k$ terms ($\triangle$ the slide writes “ $\beta_k\lambda_k$ ”; the asset subscript i is missing).
- **One price, many betas**: λ_k is a market-wide price. If $\lambda_k \approx 0$, that risk earns nothing and isn’t priced.

- **Assumptions:**

- $\alpha_i = 0$ for all i (the model is correctly specified).
- Loadings are constant.
- Factors are traded (for $\lambda = E[F]$).
- Otherwise, expected returns deviate from the hyperplane by α_i .

- **Interpretation of signs:**

- $\beta_{ik}\lambda_k > 0$: the exposure *raises* the required return.

- A **negative loading** on a factor with $\lambda_k > 0$ *lowers* the required return. The asset **hedges** that risk, and “hedges are worth paying for” [S14].
- **Traps:**
 - Adding R_f to an *excess*-return equation twice.
 - Mixing total and excess: $E[\tilde{R}_i]$ (total) = $R_f + \dots$, while $E[\tilde{R}_i - R_f] = \dots$ (no R_f).
 - Using $E[R_M]$ instead of $E[R_M - R_f]$ as λ_M .
- **Connections:** W3 SML; W6 SDF (the premia λ come from covariances with the SDF).

1.6 Fama-MacBeth two-pass regression [S12]

- **Procedure:**
 1. **First pass (time series):** for each asset i , regress its returns on the factor over $t = 1 \dots T$: $\tilde{R}_{it} - R_f = \alpha_i + \beta_i \tilde{F}_t + \tilde{\varepsilon}_{it}$. This gives one loading β_i per asset.
 2. **Second pass (cross-section):** at each date t , regress the cross-section of returns on the estimated β s: $\tilde{R}_{it} = \tilde{\gamma}_{0t} + \tilde{\gamma}_{1t} \hat{\beta}_i + \tilde{\eta}_{it}$. This gives one slope $\tilde{\gamma}_{1t}$ per period.
 3. **Average the slopes:** $\hat{\lambda} = \frac{1}{T} \sum_{t=1}^T \tilde{\gamma}_{1t}$. “Their spread over time gives the standard error and t-stat on λ .”

- **Standard error [Beyond slides formula]:**

$$SE(\hat{\lambda}) = \frac{s(\tilde{\gamma}_{1t})}{\sqrt{T}}, \quad s^2 = \frac{1}{T-1} \sum_t (\tilde{\gamma}_{1t} - \hat{\lambda})^2, \quad t = \hat{\lambda}/SE.$$

- Same for $\hat{\gamma}_0 = \frac{1}{T} \sum \tilde{\gamma}_{0t}$.
- With K factors, the second pass has K slopes per date.

- **Reading λ :**

- λ is the slope of the $E[R]$ -vs- β line: the extra expected return per unit of exposure.
- $\lambda > 0$ means the risk is priced; $\lambda \approx 0$ means the exposure earns nothing.
- The **key contribution** of Fama-MacBeth: the premium is the mean of T cross-sectional slopes, and their variation over time delivers the standard error. This automatically accounts for cross-sectional correlation of the η s.

- **Assumptions / caveats [Beyond slides]:**

- **Errors-in-variables:** $\hat{\beta}_i$ are estimated, not true, which biases $\hat{\lambda}$ toward 0. The usual fixes are portfolios instead of single stocks, and the Shanken (1992) correction to the SEs.
- Loadings are assumed constant, or rolling.
- Fama-MacBeth SEs ignore serial correlation in γ_t .
- **Slide inconsistency:** the first pass uses **excess** returns, but the second pass is written with **raw** $\tilde{R}_{it}$.
 - With raw returns, γ_0 should be $\approx R_f$.
 - With excess returns (more common), the model predicts $\gamma_0 = 0$. A significant $\hat{\gamma}_0 \neq 0$ then signals a zero-beta-rate or misspecification problem.
 - Be consistent.
- **Worked example [Beyond slides]** (4 assets with $\hat{\beta} = (0.5, 0.8, 1.2, 1.5)$, 3 months, excess returns in %, chosen to fit exactly for hand calculation):

Month	Returns (R_1, R_2, R_3, R_4)	$\tilde{\gamma}_{0t}$	$\tilde{\gamma}_{1t}$ (slope = $\Delta R / \Delta \beta$)
1	(0.70, 1.00, 1.40, 1.70)	0.2	$(1.70 - 0.70) / (1.5 - 0.5) = \mathbf{1.0}$

Month	Returns (R_1, R_2, R_3, R_4)	$\tilde{y}_{0t}$	$\tilde{y}_{1t}$ (slope = $\Delta R / \Delta \beta$)
2	(-0.15, -0.30, -0.50, -0.65)	0.1	$(-0.65 + 0.15)/1.0 =$ -0.5
3	(1.00, 1.42, 1.98, 2.40)	0.3	$(2.40 - 1.00)/1.0 =$ 1.4

- $\hat{\lambda} = (1.0 - 0.5 + 1.4)/3 = \mathbf{0.633\% \text{ per month}}$.
- Deviations (0.367, -1.133, 0.767), so $s^2 = (0.1344 + 1.2844 + 0.5878)/2 = 1.0033$ and $s = 1.0017$.
- $SE = 1.0017/\sqrt{3} = 0.578$, so $t = 1.10$: not significant with $T = 3$.
- $\hat{y}_0 = 0.2\%$, which is $\neq 0$ in excess returns and would be evidence against the model if significant.
- **Interpretation:** monthly $\hat{\lambda} = 0.633\%$ is $\approx 7.6\%$ a year. The big month-to-month swings in y_{1t} (up and down markets) are why you need long T .
- **Traps:**
 - Using T instead of $\sqrt{T}$ in the SE.
 - Running the cross-section on *average* returns only once. That gives the point estimate but no FM standard error.
 - Confusing $\tilde{y}_{1t}$ with the FF size loading γ_i .
- **Implementation (tested):**

```
def fama_macbeth(Rex, F):
    """Rex: T x N excess returns, F: T x K factors -> lambda (incl. gamma0), SE, t."""
    T, N = Rex.shape
    betas = np.array([ts_regress(Rex[:, i], F)[1] for i in range(N)]) # pass 1
    Xc = np.column_stack([np.ones(N), betas])
    gam = np.array([np.linalg.lstsq(Xc, Rex[t], rcond=None)[0] for t in range(T)]) # pass 2
    lam = gam.mean(0); se = gam.std(0, ddof=1) / np.sqrt(T) # step 3
    return lam, se, lam / se
```

Pitfall: using single stocks leads to severe errors-in-variables. Sort into portfolios first (e.g. 25 size $\times$ B/M).

1.7 Factors as traded long-short portfolios; indicative premia [S13]

- **Definition:**
 - A factor is a **self-financing long-short portfolio**: long the assets with a trait, short those without.
 - Because you can actually hold it, its risk premium is just its average return, $\lambda = E[f]$.
 - Long-short means “both cost almost nothing to hold” [S17] (zero net investment, ignoring margin and short costs [Beyond slides]).

Factor	Long – short portfolio	What it rewards	Premium $E[f]^*$
Market (MKT)	All equities – T-bills	Bearing equity risk	$\sim 6\%$
Size (SMB)	Small caps – large caps	Small-firm risk	$\sim 2\%$
Value (HML)	Cheap – expensive (B/M)	Value over growth	$\sim 3\%$
Momentum (WML)	Past winners – past losers	Trend persistence	$\sim 7\%$

- *Indicative long-run US averages (annual); they vary by sample and market.
- The **Fama-French 5-factor** model adds **profitability (RMW, robust minus weak)** and **investment (CMA, conservative minus aggressive)**. The expansions are Beyond slides; the slides give only RMW/CMA.

- **▲ Naming:** momentum is WML here, UMD in the Carhart slides, and “Mom” in the Ken French library. All are the same concept.
- **Traps:**
 - These premia are *annual* and *indicative*. Monthly $\approx$ annual/12.
 - “Premium” is the average *return of the long-short portfolio*, not a characteristic value.
 - MKT is $\text{Mkt} - R_f$, not the total market return.

1.8 Worked example: a two-factor model [S14]

- **Steps:**
 1. Write the return: $\tilde{R}_i - R_f = \alpha_i + \beta_{i1} \tilde{F}_1 + \beta_{i2} \tilde{F}_2 + \tilde{\varepsilon}_i$, with $\tilde{F}_1$ = market excess return and $\tilde{F}_2$ = value (HML).
 2. Estimate the loadings: one time-series regression of $\tilde{R}_i - R_f$ on $\tilde{F}_1, \tilde{F}_2$ gives slopes β_{i1} and β_{i2} .
 3. Price it: $E[\tilde{R}_i] = R_f + \beta_{i1} \lambda_1 + \beta_{i2} \lambda_2$, with $\lambda_k = E[\tilde{F}_k]$.
- **Numbers (slide):** $R_f = 2\%$; $\beta_{i1} = 1.2$ with $\lambda_1 = 5\%$; $\beta_{i2} = -0.4$ with $\lambda_2 = 3\%$.

$$E[\tilde{R}_i] = 2 + 1.2(5) + (-0.4)(3) = 2 + 6 - 1.2 = \mathbf{6.8\%} \checkmark$$

- **Interpretation:**
 - The **negative** value loading *lowers* the required return by 1.2 points. The asset behaves like growth, i.e. it **hedges value risk**, and “hedges are worth paying for”.
 - Compared with CAPM using the same market β : $2 + 6 = 8\%$. The value factor changes the required return by -1.2% .
- **Variant [Beyond slides]:** with $\beta_{i2} = +0.4$ instead, $E[\tilde{R}_i] = 9.2\%$.
- **Traps:**
 - Sign errors on negative loadings.
 - Adding R_f when the question asks for the expected *excess* return (which would be 4.8%).
- **In-class quiz Q1-Q3 [S15]:** the general factor form, loadings versus premia, and the two-factor example. The slides show a title only.

1.9 Size and value anomalies → the Fama-French three-factor model [S16-S18]

- **Motivation [S17]: two patterns the CAPM could never absorb:**
 - **Size effect:** small-capitalisation firms historically earned more than their β justified. Banz (1981, *JFE* 9(1)) documented it, and it “is still argued over”.
 - **Value effect:** high book-to-market firms beat low book-to-market firms, again beyond what β explains (“value beats growth”).
 - **The pragmatic move:** rather than explain the patterns away, build portfolios that capture them and use those portfolios as factors (Fama & French 1993, *JFE* 33(1)).
 - SMB = **small minus big**; HML = **high minus low** book-to-market. Both are long-short.
- **Model [S18]:**

$$\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \gamma_i \text{SMB} + \delta_i \text{HML} + \tilde{\varepsilon}_i$$

$$E[\tilde{R}_i] = R_f + \beta_i \cdot \text{MRP} + \gamma_i E[\text{SMB}] + \delta_i E[\text{HML}]$$
 - The expectation form assumes $\alpha_i = 0$ (the slide leaves this implicit).
- **Reading the loadings:**

- β (**market**): exactly as in the CAPM, how much of the market's move this asset carries. Note that the FF3 β is a *multiple*-regression slope, so it generally differs from the CAPM β [Beyond slides].
- γ (**size**): positive means the asset *behaves like* a small-cap stock, "whatever its actual size".
- δ (**value**): positive means it behaves like a value stock; negative means it behaves like growth.
- **Assumptions:** SMB and HML are traded, so their premia are their means. Loadings are constant. $\alpha_i = 0$ under the model.
- **Worked example [Beyond slides, using the [S13] premia]:**
 - Inputs: $R_f = 2\%$, $MRP = 6\%$, $E[SMB] = 2\%$, $E[HML] = 3\%$; stock with $\beta = 1.1$, $\gamma = 0.5$, $\delta = -0.3$.
 - $E[\tilde{R}_i] = 2 + 1.1(6) + 0.5(2) + (-0.3)(3) = 2 + 6.6 + 1.0 - 0.9 = \mathbf{8.7\%}$.
 - CAPM with the same β would give 8.6%. The size tilt adds +1.0 and the growth tilt subtracts 0.9.
- **Interpretation:**
 - A stock with large positive γ and δ (small value) has a high required return.
 - A fund's CAPM alpha can be "explained away" if it simply tilts toward small or value stocks.
- **Traps:**
 - Reading $\gamma > 0$ as "the firm is small". It means the firm *co-moves* with small stocks.
 - Using the CAPM β inside FF3.
 - Forgetting the MRP multiplies β while $E[SMB]$ multiplies γ .
- **Connections:** the W3 CAPM anomalies; W5 factor exposures B .

1.10 Data: the Ken French Data Library [S19-S20]

- **Source:** mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html (free, public).
- **What you get:**
 - Mkt-RF, SMB, HML (3-factor); + RMW, CMA (5-factor); Momentum (Mom / WML).
 - Sorted and industry portfolios.
 - US, developed and emerging regions.
- **Frequency and history:**
 - Daily, weekly, monthly, annual.
 - US history from **July 1926**; developed and global from **1990**.
 - Refreshed every month; same columns at every frequency.
- **Format and gotchas:**
 - CSV inside a .zip, with monthly rows first and an **annual block below**.
 - Returns are **in percent**, so $\div 100$ before use.
 - **-99.99 marks a missing month.**
 - **Mkt-RF is already excess of RF.**
- **Access:**
 - Browser: pick a dataset $\rightarrow$ download the CSV $\rightarrow$ unzip.
 - Python: `web.DataReader('F-F_Research_Data_5_Factors_2x3', 'famafrch')` (pandas-datareader).
- **Raw data [S20]** (FF5, monthly, %):

Date	Mkt-RF	SMB	HML	RMW	CMA	RF
2020-01	-0.11	-3.10	-3.12	1.30	0.60	0.13
2020-02	-8.13	1.09	-3.86	2.20	1.90	0.12
2020-03	-13.39	-5.16	-14.13	4.60	4.90	0.12
2020-04	13.65	2.79	-1.20	-2.10	-1.50	0.00

Date	Mkt-RF	SMB	HML	RMW	CMA	RF
2020-05	5.58	2.49	-4.99	-0.50	-1.90	0.01
2020-06	2.46	1.86	-2.02	-0.80	0.60	0.01
2020-07	5.77	-2.30	-1.50	1.10	0.90	0.01
2020-08	7.63	-0.99	-2.95	1.30	-0.60	0.01

- The slide calls these “illustrative monthly rows”; March 2020 (COVID crash) is highlighted.
- Worked examples [Beyond slides]:**
 - Total market return, March 2020** = $\text{Mkt-RF} + \text{RF} = -13.39 + 0.12 = -13.27\%$, i.e. -0.1327 in decimals.
 - Cumulative total market return, Jan-Aug 2020** = $\prod (1 + (\text{MktRF}_t + \text{RF}_t)/100) - 1 = +11.73\%$. The simple sum of total returns (Mkt-RF sum 13.46 + RF sum 0.41) is 13.87%, which is wrong for a holding-period return.
 - HML over the 8 months**: mean -4.22% /month, compounded **-29.7%**. This was the 2020 “value crash”, when growth massively outperformed.
 - RMW and CMA were positive in March 2020**: profitable, conservative firms held up in the crash.
- Traps:**
 - Forgetting $\div 100$.
 - Subtracting RF from Mkt-RF again.
 - Summing monthly returns instead of compounding.
 - Treating the annual block as extra monthly rows.
 - Leaving -99.99 in the data.
- Implementation:**

```
import pandas as pd, pandas_datareader.data as web
ff = web.DataReader('F-F_Research_Data_5_Factors_2x3', 'famafrench', start='1990')[0] # [0] =
    ~ monthly table
ff = ff.replace(-99.99, float('nan')) / 100 # % -> decimal, flag missing
mkt_total = ff['Mkt-RF'] + ff['RF'] # total market return
cum = (1 + mkt_total).prod() - 1 # compound, don't sum
```

Pitfall: the reader returns a dict of tables (0 = monthly, 1 = annual). Index 1 is the annual block.

1.11 How the factors are built: Mkt-RF, SMB, HML [S21-S23]

- Market [S21]:**
 - Take every US stock: all common shares on NYSE, AMEX and NASDAQ (the whole CRSP universe).
 - Weight by size: value-weight each stock by market cap. This gives R_m .
 - Subtract cash: the one-month Treasury bill, R_f .
 - Result: **Mkt-RF** = $\tilde{R}_m - R_f$. This is the CAPM market premium, “the reward for holding equities instead of cash”; every other factor is measured on top of it.
- 2x3 sort (shared by SMB and HML):**
 - Breakpoints: size at the **NYSE median**; B/M at the **30th and 70th NYSE percentiles**.
 - This gives six portfolios: S/L, S/M, S/H, B/L, B/M, B/H (Small/Big $\times$ Low/Neutral/High B/M).
 - Value-weighted** inside each portfolio.
 - Rebalancing: **re-sorted once a year, end of June; held 12 months**.
- SMB [S22]:**

$$\text{SMB} = \frac{1}{3}(S/L + S/M + S/H) - \frac{1}{3}(B/L + B/M + B/H)$$
 - The average of the three small portfolios minus the average of the three big portfolios.
 - It isolates the size premium “while holding value exposure fixed across both legs”.

- **HML [S23]:**

$$\text{HML} = \frac{1}{2}(S/H + B/H) - \frac{1}{2}(S/L + B/L)$$

- The average of the two high-B/M (value) portfolios minus the average of the two low-B/M (growth) portfolios. The neutral S/M and B/M are **not used**.
- It isolates the value premium “while holding size exposure fixed across both legs”.
- **Why NYSE breakpoints [Beyond slides]:** NASDAQ/AMEX add many tiny firms. NYSE breakpoints stop the “big” group from being just a handful of mega-caps and the small group from being dominated by micro-caps.
- **Why end of June [Beyond slides]:** accounting data for fiscal year $t - 1$ (B/M uses book equity at FYE $t - 1$ over market equity in December $t - 1$) is public by June t , which avoids look-ahead bias.
- **Worked example [Beyond slides]** (monthly returns in %: S/L 1.2, S/M 1.0, S/H 1.5, B/L 0.8, B/M 0.7, B/H 1.0):
 - $\text{SMB} = \frac{1.2+1.0+1.5}{3} - \frac{0.8+0.7+1.0}{3} = 1.233 - 0.833 = \mathbf{0.40\%}$.
 - $\text{HML} = \frac{1.5+1.0}{2} - \frac{1.2+0.8}{2} = 1.25 - 1.00 = \mathbf{0.25\%}$.
- **Traps:**
 - Averaging HML over three portfolios per leg (it uses two).
 - Using the S/M and B/M portfolios in HML.
 - Equal-weighting within portfolios (FF value-weights).
 - Sorting monthly (the sort is annual).
- **Implementation:**

```
def smb_hml(p):    # p: dict of the six portfolio returns (arrays or Series)
    smb = (p["SL"] + p["SM"] + p["SH"]) / 3 - (p["BL"] + p["BM"] + p["BH"]) / 3
    hml = (p["SH"] + p["BH"]) / 2 - (p["SL"] + p["BL"]) / 2
    return smb, hml
print(smb_hml(dict(SL=1.2, SM=1.0, SH=1.5, BL=0.8, BM=0.7, BH=1.0)))    # (0.4, 0.25)
```

1.12 Reading loadings: the style map [S24]

- **Statement:**
 - Plot each of ten industries' FF3 loadings: δ (value, x-axis) against γ (size, y-axis).
 - Data: Fama-French loadings, ten industries, 2004-2013.
 - “A regression recovers the style labels nobody gave it.”
 - Quadrants: top-left small growth, top-right small value, bottom-left large growth, bottom-right large value.
- **Loadings:** the three called out on the slide are exact; the rest are read from the chart ($\approx \pm 0.03$).

Industry	δ (value)	γ (size)	Reading
HiTec	-0.54	$\approx +0.28$	strongly growth, mildly small: “the label a human would give it”
Durbl (durables)	$\approx +0.25$	+0.77	the most small-cap-like of the ten
Utils	≈ -0.07	-0.43	the most large-cap-like, otherwise style-neutral
Other	$\approx +0.56$	≈ -0.05	most value-tilted

Industry	δ (value)	γ (size)	Reading
Shops	≈ -0.03	$\approx +0.32$	small-ish, neutral value
Manuf	≈ 0.00	$\approx +0.10$	near the origin
Telcm	≈ -0.02	≈ -0.06	near the origin
Hlth	≈ -0.09	≈ -0.18	large, slight growth
Enrgy	≈ -0.11	≈ -0.27	large, slight growth
NoDur	$\approx +0.02$	≈ -0.24	large, neutral

- **Interpretation:** loadings describe **behaviour**, not characteristics. A regression on factor returns alone recovers intuitive style labels, which is evidence that the factors capture real common variation.
- **Traps:**
 - Reading a small $|\delta|$ as “no value exposure” without checking its t-stat.
 - Loadings are sample-dependent (2004–2013 here).
- **Connection:** quiz Q4–Q6 [S26] covers Fama–French, the style map, and what α measures.

1.13 Momentum: the pattern and its construction [S27–S28, S30]

- **Pattern [S28]:**
 - Sort on the past **3–12 months**, buy the winners, short the losers, “and the spread keeps earning”.
 - “Winners keep winning, for about a year” (months 1–12 after the sort).
 - **It reverses:** beyond a year the sign flips. Returns over 3–5 years run against the previous 3–5 years (**long-run reversal**, De Bondt & Thaler).
 - **Why it is awkward:** it is a pure price pattern, with no accounting data, no valuation, “nothing about the firm”. Even **weak-form efficiency** (prices reflect past prices) says it should not work.
 - Source: Jegadeesh & Titman (1993), *JF* 48(1).
 -  The schematic’s x-axis runs only to ~36 months but shades the band from month 24 to 36 as “years 3–5”; strictly, that band is year 3. The text statement (3–5-year reversal) is the one to quote.
- **Original construction [S30] (Jegadeesh–Titman):**
 - Rank every stock by its past return into deciles D1 (lowest, losers) to D10 (highest, winners).
 - **Momentum = top-decile winners – bottom-decile losers:** long D10, short D1, **equal-weighted**, held 3–12 months.
 - **Formation:** prior 3–12 months (classic: 6). **Holding:** next 3–12 months (classic: 6).
 - Original method: one sort on past return, **no size split**; deciles, equal-weighted.
- **Momentum variants across the slides **:

Version	Sort	Look-back	Legs	Weighting	Rebalance
J&T original [S30]	deciles of all stocks	3–12 m (6)	D10 – D1	equal	hold 3–12 m (6)
UMD in Carhart [S25, S29]	past-year return	“past year”	winners – losers	—	monthly
Class stand-in [S29]	ten industries	past 12 m, skipping the latest month	best 3 – worst 3 industries	(equal)	monthly

Version	Sort	Look-back	Legs	Weighting	Rebalance
Ken French "Mom" [Beyond slides]	2×3 size × prior return (30/70 NYSE)	months $t - 12$ to $t - 2$	$\frac{1}{2}(S/Hi+B/Hi)$ — $\frac{1}{2}(S/Lo+B/Lo)$	value	monthly

- **Why skip the latest month [Beyond slides]:** one-month returns show **short-term reversal** (bid-ask bounce, liquidity). Including month $t - 1$ contaminates the momentum signal. This is the "12-1" convention.
- **Traps:**
 - Confusing momentum (3-12 months, continuation) with reversal (1 month and 3-5 years, mean-reverting).
 - Including the most recent month in the signal.
 - Look-ahead: the ranking must use only returns known before the holding period.
- **Implementation (class stand-in, tested):**

```
import numpy as np, pandas as pd
def umd_standin(ind, n_leg=3, look=12, skip=1):
    """ind: T x 10 DataFrame of monthly industry returns (decimal). Long best n_leg, short
    ~ worst n_leg
    on compounded return over months t-look .. t-1-skip (i.e. 12-1)."""
    past = (1 + ind).rolling(look - skip).apply(np.prod, raw=True).shift(skip + 1) - 1
    out = pd.Series(np.nan, index=ind.index)
    for t in ind.index[look:]:
        r = past.loc[t].dropna().sort_values()
        if len(r) >= 2 * n_leg:
            out[t] = ind.loc[t, r.index[-n_leg:]].mean() - ind.loc[t, r.index[:n_leg]].mean()
    return out
```

Pitfall: the .shift(skip+1) makes the signal at t use only returns up to t - 2. Drop it and you get look-ahead bias. The coding session asks you to vary look, skip and n_leg to see how fragile the result is.

1.14 Carhart four-factor model, UMD loadings, and what α measures [S25, S29, S31]

- **Model:**

$$\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \gamma_i \text{SMB} + \delta_i \text{HML} + \zeta_i \text{UMD} + \tilde{\varepsilon}_i$$
 - UMD = **up minus down**: long past winners, short past losers, rebalanced monthly.
 - "Three factors + one = the Carhart four": start with FF3 (market, size, value, "three priced factors that already beat CAPM"), add momentum (the pattern the three left on the table), and get the **four-factor benchmark**.
 - "The standard yardstick for funds. Whatever α survives all four factors is the return exposures can't explain."
- **Why it was added [S29]:** Carhart (1997, *JF* 52(1)) showed that mutual-fund "skill" (performance persistence) largely vanished once momentum exposure was controlled for. "Much of what looked like skill was momentum." [S25]
- **Awkward status:** "Everyone uses it. Nobody agrees on what risk it represents, or whether it is a risk at all." There is no agreed risk story, yet it is universally used as a benchmark.
- **What α measures:**
 - α_i is the average excess return **not** attributable to the factor exposures, i.e. skill or mispricing *relative to the chosen benchmark model*.
 - α depends on the model: CAPM $\alpha \neq$ FF3 $\alpha \neq$ Carhart α .
 - The coding session: run CAPM, FF3 and Carhart on each industry; "watch α shrink and R^2 rise, and check which α survives".
- **Momentum loadings, ten industries [S31] (ζ):**

Energy	Utils	Hlth	Telcm	NoDur	HiTec	Manuf	Shops	Other	Durbl
+0.53	+0.26	+0.10	+0.04	+0.01	-0.05	-0.06	-0.09	-0.21	-0.58

- **Energy +0.53** “tends to be in the winner basket when anything is trending”, so it moves with the factor.
- **Durables -0.58** is “the contrarian of the ten”. Adding the UMD column **pulled its α from -4.8% to -2.7%**.
- **Why Durables’ α rose [Beyond slides]:**
 - FF3 α absorbs the omitted term $\zeta E[\text{UMD}]$: $\alpha_{FF3} \approx \alpha_{C4} + \zeta E[\text{UMD}]$.
 - With $\zeta = -0.58$ and $E[\text{UMD}] > 0$, omitting UMD makes α look *more negative*.
 - Back-of-envelope: $E[\text{UMD}] \approx \frac{-2.7 - (-4.8)}{0.58} = \frac{2.1}{0.58} \approx 3.6\%$ a year for the classroom series. This is approximate, because the other loadings also shift when a column is added.
- **Worked example [Beyond slides]: “skill was momentum”:**
 - A fund earns an average excess return of 9%/yr. Market premium 6%, $E[\text{SMB}] = 2\%$, $E[\text{HML}] = 3\%$, $E[\text{UMD}] = 7\%$.
 - CAPM ($\beta = 1.0$): $\alpha = 9 - 6 = +3.0\%$. It looks skilled.
 - Carhart ($\beta = 1.0$, $\gamma = 0.3$, $\delta = 0.2$, $\zeta = 0.4$): predicted $6 + 0.6 + 0.6 + 2.8 = 10.0\%$, so $\alpha = 9 - 10 = -1.0\%$.
 - All of the “alpha” was factor exposure, mostly momentum.
- **Traps:**
 - Comparing α s estimated under different models as if they measured the same thing.
 - Assuming loadings stay fixed when a factor is added.
 - Interpreting UMD’s premium as compensation for a known risk.
- **Implementation:**

```
# y: fund/industry excess returns; ff: DataFrame with Mkt-RF, SMB, HML, Mom (decimal)
for cols in (["Mkt-RF"], ["Mkt-RF", "SMB", "HML"], ["Mkt-RF", "SMB", "HML", "Mom"]):
    a, b, t, r2 = ts_regress(y.values, ff[cols].values)
    print(len(cols), f"alpha={12*a:.2%}/yr  t={t[0]:.2f}  R2={r2:.2f}")
```

Pitfall: annualise monthly α by $\times 12$ (not compounding) when quoting “per year”. The t-stat is unchanged.

1.15 Defending against the factor zoo: four tests [S32-S36]

- **Checklist [S32]:** “You will be pitched factors for the rest of your career.”
 1. **Out of sample?** Other countries, other decades, and after the paper was published.
 2. **Why should it pay?** A story about risk or about a bias, or only a regression?
 3. **Is it already known?** Many new factors are old factors renamed, at a 0.9 correlation.
 4. **Survives costs?** Signals needing monthly turnover often earn nothing after trading.

Test 1: out-of-sample survival and multiple testing [S33]

- Growth of \$100, 2005–2026: momentum (a real signal) reaches **\$231**, while a data-mined signal peaks around when it was “found here” (~2014, ~\$155) and ends at **\$149**.
 - [Beyond slides] \$231 over 21 years $\approx 4.1\%/yr$ CAGR; \$149 $\approx 1.9\%/yr$, almost all of it before discovery.
- “Test hundreds of signals and the luckiest clears $t > 2$ by chance. With ~300 published factors, raise the bar: demand $t > 3.0$ ” (Harvey-Liu-Zhu 2016).
- Momentum was found in US data before 1993, then held in Europe, Japan and later decades. A mined signal flattens the moment you leave its sample.
- “In-sample fit is free. Out-of-sample survival is the evidence.”
- [Beyond slides] With 300 independent true-null signals tested at 5%, you expect $300 \times 0.05 = 15$ **false discoveries**. $t > 3$ corresponds to a two-sided $p \approx 0.0027$.

Test 2: a premium needs a reason [S34]

- “A factor pays only if $E[\tilde{R}_i - R_f] = \beta_i \lambda$ with $\lambda \neq 0$: a reward for risk investors avoid, or a persistent bias.”

- Chart: average excess return vs β , with a priced line of $\lambda = 6\%$ per unit β (from 2% at $\beta = 0$ to 14% at $\beta = 2$) and a flat $\lambda \approx 0$ line at $\sim 2\%$.
- Example: market and value carry stories (distress and crash risk). A random accounting ratio that happens to predict returns has none: “a red flag, not a factor”.
- “No story, no premium: a t-stat is not a mechanism.”
- $\triangle$ The chart’s y-axis says “average **excess** return” but both lines start at **2%** at $\beta = 0$. Under the stated test $E[R - R_f] = \beta\lambda$, the intercept should be **0**. The 2% is effectively R_f , i.e. the chart plots *total* return, or equivalently a non-zero intercept. Quote the formula, not the intercept.

Test 3: test what it adds (spanning) [S35]

- Regress the candidate on the factors you already own:

$$\tilde{F}_{new} = \alpha + b_1 \text{MKT} + b_2 \text{SMB} + b_3 \text{HML} + b_4 \text{UMD} + \tilde{e}.$$

- If $\alpha \approx 0$ (here $t = 0.5$), the candidate is **spanned**: “a relabel, not a discovery”.
- Example (slide): annual premium **5.2%** on its own, residual α **0.4%** after Mkt, SMB, HML, UMD. That is **-92%**, “explained by factors you already own”.
 - Check: $1 - 0.4/5.2 = 92.3\%$ ✓.
 - [Beyond slides] $t = 0.5$ implies $\text{SE}(\alpha) = 0.4/0.5 = 0.8\%$.
- “Many ‘quality’ and low-volatility factors run ~ 0.9 correlation with blends of size, value and profitability: repackaged, not new.”
- [Beyond slides] Spanning is the same regression as a fund’s performance α . A factor adds to the investment opportunity set (a higher max Sharpe ratio) **iff** its α on the existing factors is non-zero.

Test 4: gross vs net [S36]

- **Formula**: Net α = Gross α – turnover $\times$ cost.
- Waterfall (annual): gross α **6%** – bid-ask spread **3%** – impact & fees **2.5%** = net α **0.5%** ✓.
- “A signal at 200%+ monthly turnover pays the spread every month; at ~ 30 -50 bps a trade, the edge disappears.”
 - [Beyond slides] $2.0 \times 12 = 24\times$ a year, and at 30-50 bp that costs **7.2%-12%** a year, which exceeds most gross alphas.
- Example: short-horizon reversal looks strong on paper but rebalances constantly. After realistic costs its net alpha is indistinguishable from zero.
- “A backtest earns gross; only net alpha reaches the fund.”

Traps across the four tests

- Treating $t > 2$ as sufficient in a data-mined setting.
- Confusing “the factor predicts returns” with “the factor adds α beyond known factors”.
- Quoting monthly turnover as annual.
- Charging costs only once per year for a strategy that rebalances monthly.

Implementation

```
a, b, t, r2 = ts_regress(f_new, ff[["Mkt-RF", "SMB", "HML", "Mom"]].values)
spanned = abs(t[0]) < 3.0 # Harvey-Liu-Zhu hurdle for a new factor
net_alpha = gross_alpha - (monthly_turnover * 12) * cost_per_trade # e.g. 0.06 - 24*0.004
```

1.16 Quizzes and coding session [S15, S26, S37, S38]

- **In-class quizzes**: Q1-Q3 [S15] (general factor form, loadings vs premia, two-factor example); Q4-Q6 [S26] (Fama-French, style map, what α measures); Q7-Q9 [S37] (no topic given). The slides show titles only, with no questions.
- **Coding session [S38]**, “three tasks, in order of how much they will teach you”:
 1. **Run the three regressions**: CAPM, Fama-French and Carhart on each industry. Watch α shrink and R^2 rise, and check which α survives (1.14).

2. **Build your own momentum factor:** change the look-back window, the skip month and the number of industries in each leg, and see how fragile the result is (1.13).
3. **Score a portfolio five ways:** compute all five measures, then find a pair of assets the measures rank in opposite orders.
 - ⚠ The “five measures” are **not defined** in the Week 4 or Week 3 slides. They are most likely the standard performance measures (Sharpe ratio, Treynor ratio, Jensen’s α , information ratio, and M^2 or the appraisal ratio), but **I am not certain**. Check chap4.ipynb.
- **Files:** chap4.ipynb, W4_factor_explorer.html (interactive companion), W4_factor_lab.py (Gradio lab).

2. Comparison tables

2.1 Factor models covered

	CAPM (W3)	FF3 [S18]	Carhart 4 [S25, S29]	FF5 [S13 footnote]
Factors	Mkt–RF	+ SMB, HML	+ UMD	+ RMW, CMA (no UMD)
Equation (excess return)	$\alpha + \beta(R_M - R_f)$	$+\gamma \text{ SMB} + \delta \text{ HML}$	$+\zeta \text{ UMD}$	$+r \text{ RMW} + c \text{ CMA}$ [Beyond: notation]
Motivation	equilibrium theory	size and value anomalies (Banz 1981; FF 1993)	fund persistence was momentum (Carhart 1997)	profitability and investment patterns
Risk story	market risk	debated: distress, small-firm risk	none agreed	debated
Use	cost of capital, SML	style attribution, cost of equity	standard fund-performance benchmark	academic asset pricing
Weakness	fails on size, value, momentum	misses momentum	momentum crashes; ad hoc	still misses momentum; HML may become redundant [Beyond]

2.2 Estimating premia: time series vs cross-section

	Time-series mean ($\lambda = E[F]$) [S9]	Fama-MacBeth two-pass [S12]
Needs	a traded factor (a portfolio return)	$\hat{\beta}$ s for many assets; works for non-traded factors too
Estimate	$\bar{F} = \frac{1}{T} \sum F_t$	$\hat{\lambda} = \frac{1}{T} \sum_t \tilde{y}_{1t}$
SE	$s(F_t)/\sqrt{T}$	$s(\tilde{y}_{1t})/\sqrt{T}$
Tests	—	whether the cross-section pays λ per unit β ; $\gamma_0 = 0$ (excess)
Pitfalls	a short sample makes $\bar{F}$ noisy	errors-in-variables in $\hat{\beta}$; ignores serial correlation

	Time-series mean ($\lambda = E[F]$) [S9]	Fama-MacBeth two-pass [S12]
Agree when	the model is correctly specified ($\alpha_i = 0$)	

2.3 Regressions this week

Regression	Dimension	LHS	RHS	Output
First pass / loadings [S9, S12]	over time, per asset	$R_{it} - R_f$	factors F_t	$\alpha_i, \beta_{ik}, R^2$
Second pass [S12]	across assets, per date	R_{it}	$\hat{\beta}_i$	$\tilde{y}_{0t}, \tilde{y}_{1t}$
Performance / α [S25, S38]	over time, per fund	fund excess return	CAPM / FF3 / C4 factors	α (skill vs exposure)
Spanning [S35]	over time, per candidate	F_{new}	existing factors	α of the new factor

2.4 Factor construction

Factor	Sort	Breakpoints	Legs	Weighting	Rebalance
Mkt-RF [S21]	none (all CRSP common stocks)	—	market – 1-month T-bill	value	monthly
SMB [S22]	2×3 size × B/M	size: NYSE median; B/M: NYSE 30/70	avg 3 small – avg 3 big	value (within portfolios)	annually, end of June
HML [S23]	2×3 size × B/M	same	avg(S/H, B/H) – avg(S/L, B/L)	value	annually, end of June
Momentum, J&T [S30]	deciles on past return	cross-sectional deciles	D10 – D1	equal	hold 3–12 m
UMD stand-in [S29]	10 industries on 12-1 return	top/bottom 3	best 3 – worst 3	equal	monthly

2.5 The four factor-zoo tests [S32–S36]

Test	Question	Tool	Pass criterion	Slide example
1 Out of sample	Does it survive new data?	post-publication, international, later decades; multiple-testing hurdle	$t > 3.0$; performance persists	momentum \$231 vs mined \$149
2 Economic story	Why should it pay?	risk or behavioural rationale; $\lambda \neq 0$	a plausible mechanism	market/value: distress, crash risk

Test	Question	Tool	Pass criterion	Slide example
3 Already known?	Does it add anything?	spanning regression on MKT, SMB, HML, UMD	α significantly $\neq 0$	5.2% $\rightarrow$ 0.4% (t = 0.5): spanned
4 Costs	Does it survive trading?	net α = gross α – turnover $\times$ cost	net $\alpha > 0$	6% – 3% – 2.5% = 0.5%

3. Master formula sheet

K-factor model [S5, S7]

- $\tilde{R}_i - R_f = \alpha_i + \sum_{k=1}^K \beta_{ik} \tilde{F}_k + \tilde{\varepsilon}_i$: return decomposition (excess return = alpha + factor exposure + idiosyncratic).
- Assumptions: $E[\tilde{\varepsilon}_i] = 0$, $\text{Cov}(\tilde{\varepsilon}_i, \tilde{F}_k) = 0$, $\text{Cov}(\tilde{\varepsilon}_i, \tilde{\varepsilon}_j) = 0$.
- [Beyond] $\text{Var}(\tilde{R}_i) = \beta_i^\top \Omega \beta_i + \sigma_{\varepsilon_i}^2$ and $\text{Cov}(\tilde{R}_i, \tilde{R}_j) = \beta_i^\top \Omega \beta_j$.
- [Beyond] adjusted $R^2 = 1 - (1 - R^2) \frac{n-1}{n-K-1}$.

Pricing [S10-S11]

- $E[\tilde{R}_i] = R_f + \sum_k \beta_{ik} \lambda_k$: expected return = risk-free rate + loadings $\times$ premia.
- $\lambda_k = E[\tilde{F}_k]$ for traded factors; $\lambda_M = E[\tilde{R}_M - R_f] = \text{MRP}$.
- CAPM ($K = 1$): $E[\tilde{R}_i] = R_f + \beta_i \cdot \text{MRP}$.

Loadings [S9]

- One factor [Beyond]: $\hat{\beta}_i = \widehat{\text{Cov}}(R_i - R_f, F) / \widehat{\text{Var}}(F)$ and $\hat{\alpha}_i = \overline{R_i - R_f} - \hat{\beta}_i \bar{F}$.

Fama-MacBeth [S12]

- Pass 1: $\tilde{R}_{it} - R_f = \alpha_i + \beta_i \tilde{F}_t + \tilde{\varepsilon}_{it}$.
- Pass 2: $\tilde{R}_{it} = \tilde{\gamma}_{0t} + \tilde{\gamma}_{1t} \hat{\beta}_i + \tilde{\eta}_{it}$.
- $\hat{\lambda} = \frac{1}{T} \sum_t \tilde{\gamma}_{1t}$.
- [Beyond] $\text{SE} = s(\tilde{\gamma}_{1t}) / \sqrt{T}$ and $t = \hat{\lambda} / \text{SE}$.

FF3 and Carhart [S18, S25, S29]

- FF3: $\tilde{R}_i - R_f = \alpha_i + \beta_i(\tilde{R}_M - R_f) + \gamma_i \text{SMB} + \delta_i \text{HML} + \tilde{\varepsilon}_i$, with $E[\tilde{R}_i] = R_f + \beta_i \text{MRP} + \gamma_i E[\text{SMB}] + \delta_i E[\text{HML}]$.
- Carhart: add ζ_i UMD.

Factor construction [S21-S23, S30]

- Mkt–RF = $\tilde{R}_m - R_f$ (value-weighted CRSP minus the 1-month T-bill).
- $\text{SMB} = \frac{1}{3}(S/L + S/M + S/H) - \frac{1}{3}(B/L + B/M + B/H)$.
- $\text{HML} = \frac{1}{2}(S/H + B/H) - \frac{1}{2}(S/L + B/L)$.
- Momentum (J&T) = D10 – D1 (equal-weighted; formation and holding 3–12 m).

Factor-zoo tests [S33-S36]

- New-factor hurdle: $t > 3.0$.
- Priced iff $E[\tilde{R}_i - R_f] = \beta_i \lambda$ with $\lambda \neq 0$.
- Spanning: $\tilde{F}_{\text{new}} = \alpha + b_1 \text{MKT} + b_2 \text{SMB} + b_3 \text{HML} + b_4 \text{UMD} + \tilde{\varepsilon}$; spanned if $\alpha \approx 0$.
- Net α = gross α – turnover $\times$ cost.

Data conversions [S19-S20]

- decimal = % / 100.
- Total market = (Mkt–RF) + RF.

- Cumulative = $\prod(1 + r_t) - 1$.
-

4. Glossary

- **Alpha** (α_i): the mean excess return not explained by factor loadings; zero if the model prices the asset; model-dependent.
 - **Anomaly**: an average-return pattern a model (here the CAPM) fails to explain (size, value, momentum).
 - **B/M (book-to-market)**: book equity / market equity. High = value, low = growth.
 - **Breakpoints (NYSE)**: sort cutoffs computed from NYSE stocks only (size median; B/M 30th/70th percentiles).
 - **Carhart four-factor model**: FF3 + UMD; the standard fund benchmark.
 - **CMA**: conservative minus aggressive (investment) factor; FF5.
 - **CRSP**: Center for Research in Security Prices; US stock database (NYSE, AMEX, NASDAQ).
 - **Cross-sectional regression**: across assets at one date.
 - **Data mining / p-hacking**: testing many signals and reporting the lucky ones.
 - **Errors-in-variables**: bias from using estimated $\hat{\beta}$ as a regressor [Beyond].
 - **Excess return**: $\tilde{R}_i - R_f$.
 - **Factor**: a common source of return variation shared by many assets, here a traded long-short portfolio.
 - **Factor loading** (β_{ik}): the sensitivity of asset i to factor k ; asset-specific; estimated by time-series regression.
 - **Factor zoo**: the ~300+ published factors (Harvey-Liu-Zhu).
 - **Fama-MacBeth**: two-pass procedure; premia = average of per-period cross-sectional slopes.
 - **FF3 / FF5**: Fama-French three-factor (Mkt, SMB, HML) and five-factor (+RMW, CMA) models.
 - **Gross vs net α** : before vs after trading costs.
 - **HML**: high minus low B/M (value) factor.
 - **Hyperplane**: the K -dimensional linear surface $E[R] = R_f + \sum \beta \lambda$ on which assets should lie.
 - **Idiosyncratic risk** ($\tilde{\epsilon}_i$): firm-specific; diversifiable; unpriced.
 - **Ken French Data Library**: free source of factor and portfolio returns.
 - **Long-run reversal**: 3-5-year losers outperform 3-5-year winners (De Bondt-Thaler).
 - **Mkt-RF**: value-weighted market return minus the 1-month T-bill.
 - **Momentum (UMD / WML / Mom)**: past winners minus past losers; continuation over 3-12 months.
 - **MRP**: market risk premium $E[\tilde{R}_M - R_f]$.
 - **Out-of-sample**: data not used to discover or fit the signal.
 - **Premium** (λ_k): the price of one unit of factor- k exposure; economy-wide.
 - **Priced factor**: $\lambda_k \neq 0$.
 - R^2 : the share of return variance explained; non-decreasing in the number of regressors.
 - **RF**: one-month Treasury bill rate.
 - **RMW**: robust minus weak (profitability) factor; FF5.
 - **Self-financing (zero-cost) portfolio**: long positions funded by shorts.
 - **Short-term reversal**: 1-month return reversal; the reason for skipping the latest month.
 - **SMB**: small minus big (size) factor.
 - **Spanning test**: regress a new factor on existing ones; $\alpha \approx 0$ means nothing new.
 - **Style map**: a plot of loadings (e.g. δ vs γ) placing assets as small/large, value/growth.
 - **Time-series regression**: over time for one asset.
 - **Turnover**: fraction of the portfolio traded per period.
 - **Value vs growth**: high-B/M vs low-B/M stocks.
 - **Weak-form efficiency**: prices reflect all information in past prices, so past returns shouldn't predict future ones.
 - **2×3 sort**: independent sort into 2 size × 3 B/M groups, giving six portfolios.
-

5. Self-test

Questions

Q1 (conceptual, easy). Distinguish a factor loading β_{ik} from a factor premium λ_k : who “owns” each, how each is estimated, and which one is the same for every asset. Why is $\lambda_k = E[\tilde{F}_k]$ for a traded factor?

Q2 (numerical, easy). $R_f = 3\%$. An asset has market loading 0.8 (market premium 6%) and size loading 0.5 (SMB premium 2%). Find its expected return and expected excess return. What would the CAPM give with the same market loading?

Q3 (numerical, easy). This month’s six size $\times$ B/M portfolio returns are S/L 1.2%, S/M 1.0%, S/H 1.5%, B/L 0.8%, B/M 0.7%, B/H 1.0%. Compute SMB and HML. Which portfolios are *not* used in HML, and why is SMB averaged over three portfolios per leg?

Q4 (numerical). Five months of excess returns (%): factor $F = (2, -1, 3, 0, 1)$, asset $= (2.5, -1.0, 3.8, 0.3, 1.4)$. Estimate β and α by OLS. Is the α monthly or annual, and what is it annualised?

Q5 (conceptual). Why does in-sample R^2 never fall when a factor is added? A CAPM regression has $R^2 = 0.30$ and a 4-factor model $R^2 = 0.33$ on $n = 60$ months. Compare adjusted R^2 and interpret.

Q6 (numerical). Using the Ken French rows in 1.10: (a) the total market return in March 2020; (b) the cumulative total market return from Jan to Aug 2020; (c) the mistakes you’d make by summing, or by forgetting $\div 100$.

Q7 (numerical). A fund’s average excess return is 9%/yr. Premia: MRP 6%, SMB 2%, HML 3%, UMD 7%. (a) CAPM α with $\beta = 1$. (b) Carhart α with loadings (1.0, 0.3, 0.2, 0.4). Interpret the change.

Q8 (numerical, Fama-MacBeth). Four assets with $\hat{\beta} = (0.5, 0.8, 1.2, 1.5)$. Monthly excess returns: month 1 (0.70, 1.00, 1.40, 1.70); month 2 (−0.15, −0.30, −0.50, −0.65); month 3 (1.00, 1.42, 1.98, 2.40). Compute $\tilde{y}_{1t}$, $\hat{\lambda}$, its SE and t-stat, and $\hat{y}_0$. Is the factor priced? What does $\hat{y}_0$ tell you?

Q9 (conceptual). Explain the construction of momentum per Jegadeesh–Titman and per the classroom stand-in. Why skip the most recent month? Why is momentum “awkward” for market efficiency, and what happens at 3–5-year horizons?

Q10 (numerical, spanning and multiple testing). A new “quality” factor earns 5.2%/yr. Regressed on MKT, SMB, HML and UMD, its α is 0.4%/yr with $t = 0.5$. (a) What fraction of the premium is explained? (b) What is $SE(\alpha)$? (c) Is it a new factor? (d) If 300 useless signals are tested at the 5% level, how many “discoveries” do you expect, and what hurdle does Harvey–Liu–Zhu propose?

Q11 (numerical, costs). A signal has gross α of 6%/yr and turnover of 200% per month. Trading costs 40 bp per unit traded. Compute the annual cost and net α . Reconcile with the slide’s waterfall (6% − 3% − 2.5%).

Q12 (hard, synthesis). Durables has momentum loading $\zeta = -0.58$, and its α moves from −4.8% (FF3) to −2.7% (Carhart). (a) Explain the direction of the change using an omitted-variable argument. (b) Back out the implied $E[UMD]$, stating the assumption. (c) Show that for any traded factor included in the model, the pricing equation implies $\lambda_k = E[\tilde{F}_k]$. (d) Why must a long-only fund with $\beta > 0$ on UMD expect a positive UMD contribution in its benchmark return?

Solutions

S1.

- β_{ik} **belongs to the asset**: it is how much of factor k the asset carries, estimated by a **time-series regression** of the asset’s excess return on the factors.

- λ_k **belongs to the economy**: it is the price per unit of exposure, **the same for every asset**, observed as the factor's mean (traded) or estimated cross-sectionally (Fama-MacBeth).
- For a traded factor, apply $E[R - R_f] = \sum_j \beta_j \lambda_j$ to the factor portfolio itself. Its loading is 1 on itself and 0 on the others, and $\alpha = 0$, so $E[\tilde{F}_k] = \lambda_k$.

S2.

- $E[R] = 3 + 0.8(6) + 0.5(2) = 3 + 4.8 + 1.0 = \mathbf{8.8\%}$.
- Excess return = **5.8%**.
- CAPM: $3 + 4.8 = \mathbf{7.8\%}$. The size tilt adds 1.0%.

S3.

- $\text{SMB} = \frac{3.7}{3} - \frac{2.5}{3} = 1.233 - 0.833 = \mathbf{0.40\%}$.
- $\text{HML} = \frac{2.5}{2} - \frac{2.0}{2} = 1.25 - 1.00 = \mathbf{0.25\%}$.
- The neutral S/M and B/M portfolios are unused in HML: it contrasts only the extremes (top and bottom 30% of B/M).
- SMB averages across all three B/M groups so that each leg has the same value exposure. The size premium is isolated "holding value fixed". (HML averages over both size groups for the same reason.)

S4.

- Deviations: $F : (1, -2, 2, -1, 0)$ and $R : (1.1, -2.4, 2.4, -1.1, 0)$.
- $\widehat{\text{Cov}} = 11.8/4 = 2.95$ and $\widehat{\text{Var}} = 10/4 = 2.5$, so $\hat{\beta} = 1.18$.
- $\hat{\alpha} = 1.4 - 1.18(1.0) = \mathbf{0.22\% \text{ per month}}$ (the data are monthly), $\approx \mathbf{2.64\% \text{ per year}}$ ($\times 12$).

S5.

- OLS minimises the SSR. The larger model can always reproduce the smaller one by setting the new coefficients to 0, so $\text{SSR}_{\text{new}} \leq \text{SSR}_{\text{old}}$ and R^2 can't fall, even on pure noise [S5].
- Adjusted: CAPM $1 - 0.70 \cdot \frac{59}{58} = \mathbf{0.288}$; four-factor $1 - 0.67 \cdot \frac{59}{55} = \mathbf{0.281}$.
- Adjusted R^2 falls, so the extra three factors don't earn their degrees of freedom here.

S6.

- (a) $-13.39 + 0.12 = \mathbf{-13.27\%}$.
- (b) $\prod_t (1 + (\text{MktRF}_t + \text{RF}_t)/100) - 1 = \mathbf{+11.73\%}$.
- (c) Summing gives 13.87% (it ignores compounding). Forgetting $\div 100$ makes $\prod(1 + r)$ explode, e.g. $1 + (-13.27)$ is negative. Also don't subtract RF again, because Mkt-RF is already excess.

S7.

- (a) CAPM $\alpha = 9 - 1.0(6) = \mathbf{+3.0\%}$.
- (b) Carhart predicted $= 6 + 0.3(2) + 0.2(3) + 0.4(7) = 6 + 0.6 + 0.6 + 2.8 = 10.0\%$, so $\alpha = \mathbf{-1.0\%}$.
- The apparent skill came from small, value and especially momentum tilts (2.8%). "Much of what looked like skill was momentum." The fund underperforms its factor benchmark.

S8.

- Each month fits exactly, so the slopes are:
 - $\tilde{\gamma}_{11} = (1.70 - 0.70)/(1.5 - 0.5) = 1.0$.
 - $\tilde{\gamma}_{12} = (-0.65 + 0.15)/1.0 = -0.5$.
 - $\tilde{\gamma}_{13} = (2.40 - 1.00)/1.0 = 1.4$.
- $\hat{\lambda} = 1.9/3 = \mathbf{0.633\%/month}$.
- $s = \sqrt{2.0067/2} = 1.0017$, so $\text{SE} = 1.0017/\sqrt{3} = \mathbf{0.578}$ and $\mathbf{t = 1.10}$.
- Not significantly priced (need $|t| \geq 2$, and with $T = 3$ even more).
- Intercepts are 0.2, 0.1 and 0.3, so $\hat{\gamma}_0 = \mathbf{0.2\%}$. With excess returns the model predicts $\gamma_0 = 0$; a significant γ_0 would mean zero-beta assets earn a premium, i.e. misspecification.

S9.

- **J&T**: rank all stocks on their prior 3-12-month return (classic 6) into deciles; go long D10 and short D1, equal-weighted; hold 3-12 months (classic 6); no size split.
- **Stand-in**: rank ten industries on the past-12-month return skipping the latest month; long the best 3, short the worst 3; monthly.
- **Skip the latest month** because one-month returns reverse (microstructure, liquidity), which would contaminate the signal.
- **Awkward**: it is a pure price pattern that violates weak-form efficiency, and there is no agreed risk story.
- At **3-5 years** it reverses (De Bondt–Thaler long-run reversal).

S10.

- (a) $1 - 0.4/5.2 = \mathbf{92.3\%}$ is explained.
- (b) $SE = 0.4/0.5 = \mathbf{0.8\%}$.
- (c) **No**. It is spanned by existing factors: “a relabel, not a discovery” (quality/low-vol run ~ 0.9 correlation with size, value and profitability blends).
- (d) $300 \times 0.05 = \mathbf{15}$ false discoveries. Harvey–Liu–Zhu (2016) propose $t > 3.0$.

S11.

- Annual turnover = $2.0 \times 12 = 24$, so cost = $24 \times 0.004 = \mathbf{9.6\%/yr}$ and net $\alpha = 6 - 9.6 = \mathbf{-3.6\%}$.
- The slide’s waterfall uses lower realised costs (3% spread + 2.5% impact/fees = 5.5%), leaving 0.5%.
- Either way, high-turnover signals lose most or all of their gross edge.

S12.

- (a) FF3 omits UMD. Its α absorbs $\zeta E[\text{UMD}]$: $\alpha_{FF3} \approx \alpha_{C4} + \zeta E[\text{UMD}]$. With $\zeta < 0$ and $E[\text{UMD}] > 0$, the FF3 α is *more negative*. Adding UMD attributes part of the underperformance to Durables’ contrarian exposure, so α rises from -4.8% to -2.7% .
- (b) $E[\text{UMD}] \approx (\alpha_{FF3} - \alpha_{C4})/\zeta = (-4.8 + 2.7)/(-0.58) = \mathbf{3.6\%/yr}$, assuming the other loadings are unchanged when UMD is added (only approximately true).
- (c) Regress $\tilde{F}_k$ on all factors: its loading is 1 on itself and 0 on the others, with residual 0. The pricing equation then gives $E[\tilde{F}_k] - 0 = 1 \cdot \lambda_k$. (No R_f appears because $\tilde{F}_k$ is an excess / zero-cost return.) ■
- (d) Its benchmark (required) return includes $\zeta \lambda_{UMD} > 0$. So part of its raw return is compensation for (or exposure to) momentum, and only the return above that counts as α .

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title: Linear Factor Models	header
S2	Aug 2007: S&P +1.4% vs quant -6.85% (3 days); four funds, one exposure	1.1
S3	4 funds $\rightarrow$ 1 bet	1.1
S4	“One factor left too much unexplained”	1.1
S5	K-factor model and term definitions; R^2 on random numbers	1.2
S6	Act 1 title	§0

Slide	Content	Section
S7	Three pieces; ε diversifiable; CAPM $K = 1$	1.2
S8	Line / plane / hyperplane geometry	1.3
S9	$\tilde{F}$ observed, β estimated, $\lambda = E[F]$	1.4
S10	Loading = asset's, premium = economy's; pricing equation	1.5
S11	$\lambda_M = \text{MRP}$; CAPM $k = 1$; $\lambda \approx 0$ unpriced	1.5
S12	Fama-MacBeth two passes	1.6
S13	Factors as long-short portfolios; premia table; FF5 note	1.7
S14	Two-factor example (6.8%)	1.8
S15	Quiz Q1-Q3	1.8, 1.16
S16	Act 2 title	\$0
S17	Size & value effects; SMB/HML; references	1.9
S18	FF3 model and loading interpretation	1.9
S19	Ken French Data Library	1.10
S20	Raw FF5 data rows	1.10
S21	Mkt-RF construction	1.11
S22	SMB 2×3 construction	1.11
S23	HML construction	1.11
S24	Style map (HiTec, Durbl, Utils)	1.12
S25	FF + momentum column = Carhart	1.14
S26	Quiz Q4-Q6	1.12, 1.16
S27	Act 3 title	\$0
S28	Winners keep winning; reversal	1.13
S29	Carhart four-factor; status; class stand-in	1.13, 1.14
S30	J&T momentum construction	1.13
S31	Momentum loadings across industries	1.14
S32	Four questions for any new factor	1.15
S33	Test 1: out of sample, $t > 3$	1.15
S34	Test 2: economic story, $\lambda \neq 0$	1.15
S35	Test 3: spanning (5.2% $\rightarrow$ 0.4%)	1.15
S36	Test 4: net α ($6 - 3 - 2.5 = 0.5$)	1.15
S37	Quiz Q7-Q9	1.16
S38	Coding session & files	1.16

Appendix B. Flag summary (⚠)

1. **[S9] vs [S10]** λ needs “no cross-section” vs “estimate it from the cross-section”. For traded factors $\lambda = E[F]$; the cross-section (Fama-MacBeth) is for non-traded factors and for testing.
2. **[S12]** Pass 1 uses excess returns but pass 2 uses raw $\tilde{R}_{it}$. Be consistent: with excess returns the model predicts $\gamma_0 = 0$.
3. **[S34]** The “average excess return” chart has a 2% intercept at $\beta = 0$. Under $E[R - R_f] = \beta\lambda$ it should be 0; the 2% is effectively R_f (total return).
4. **[S11]** “Extra factors only add more $\beta_k \lambda_k$ terms” should read $\beta_{ik} \lambda_k$.
5. **[S5]** “k-factor model” with a last term $\beta_{ik} \tilde{F}_k$ should read $\beta_{iK} \tilde{F}_K$ (k index, K count).
6. **Act labels** S25 (“Act 3 · Adding momentum”) sits inside Act 2, and S30 (“Act 2 · How it’s built”) sits inside Act 3. Content is unaffected.
7. **Momentum definitions differ** across S25/S29 (UMD, past year, monthly), S30 (J&T deciles, EW, 3-12 m) and S29’s stand-in (industries, 12-1). WML = UMD = Mom.
8. **[S28]** The chart shades months 24-36 as “years 3-5”; the text (3-5-year reversal) is correct.
9. **[S38]** The “five measures” are not defined on the slides; my guess of the standard performance measures is unconfirmed.
10. **[S18]** The expectation form of FF3 implicitly sets $\alpha_i = 0$.
11. **Symbol clash** γ_i (FF size loading) vs $\tilde{\gamma}_{0t}, \tilde{\gamma}_{1t}$ (Fama-MacBeth coefficients).